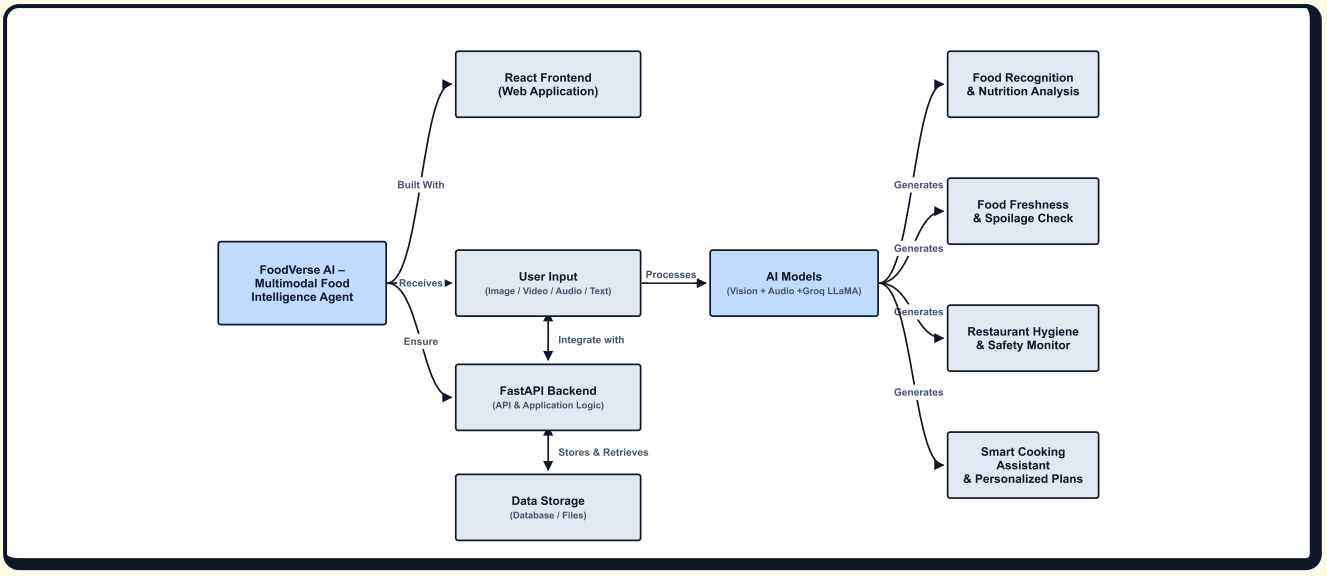
TECHNICAL ARCHITECTURE

FoodVerse AI employs a decoupled, hybrid architecture that separates the client-side user interface from the Python-based AI processing engine. This structure ensures a highly responsive UX while securely obfuscating API secrets and managing rate limits centrally.



Data Flow Responsibilities

Layer	Technology	Role in Data Flow
Frontend / Client	ReactJS, Vite, Tailwind CSS,	Handles user interactions, UI state management, and defensive input validation. Transmits formatted JSON payloads to the backend.
API Gateway	FastAPI (Python), SlowAPI	Receives client requests. Implements CORS middleware and strict rate limiting (e.g., 5 requests/min per IP) to prevent quota exhaustion.
Local Processing	Python Math/Vision Logic	Performs deterministic pre-inference calculations (Mifflin-St Jeor formulas, OpenCV pixel analysis) locally to optimize and format the prompt before hitting the cloud.
Cloud Inference	Groq API (LLaMA)	Receives the synthesized prompt from the backend. Performs heavy multimodal reasoning and natural language generation, returning actionable insights to the FastAPI gateway.